

Write a Prolog Program to Sum the Integers from 1 to n

Aim:

To sum the integers from 1 to n

Algorithm:

- Step 1: sum(N, Result) :-
- Step 2: $N > 0$, % Ensuring N is a positive integer
- Step 3: N1 is N - 1, % Decrementing N by 1
- Step 4: sum(N1, SubResult), % Recursive call to sum the integers from 1 to N1
- Step 5: Result is N + SubResult. % Adding N to the sum of integers from 1 to N1 to get the
- final result

Program:

sum(0,0). % base case: sum of 0 is 0

sum(N,Sum) :- N > 0, % recursive case: N is positive

N1 is N - 1, % decrement N

sum(N1,Sum1), % recursively compute sum of 1 to N-1

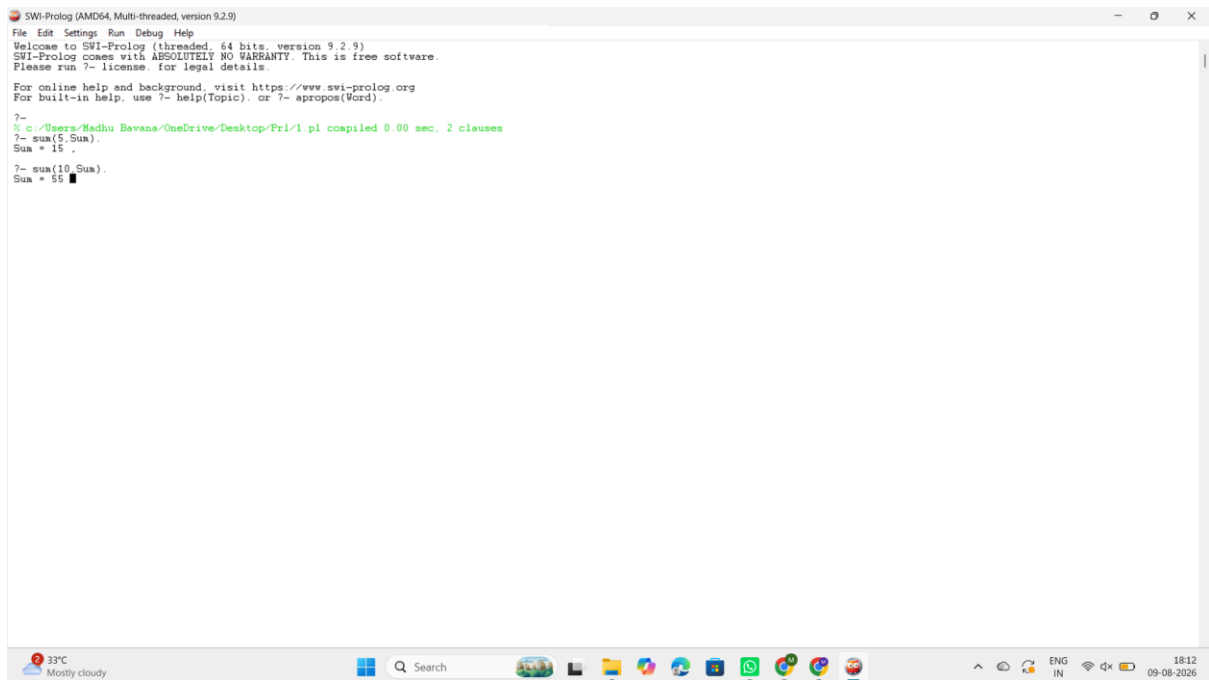
Sum is Sum1 + N. % add N to the sum of 1 to N-1

?- sum(5,Sum).

Sum = 15.

?- sum(10,Sum).

Output:



The screenshot shows the SWI-Prolog (AMD64, Multi-threaded, version 9.2.9) window. The menu bar includes File, Edit, Settings, Run, Debug, and Help. The main text area displays the following content:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/Users/Madhu Bavana/OneDrive/Desktop/Prl/1.pl compiled 0.00 sec, 2 clauses
?- sum(5, Sum).
Sum = 15.
?- sum(10, Sum).
Sum = 55
```

The Windows taskbar at the bottom shows the system clock as 18:12 on 09-08-2026, along with various system icons including network, volume, and battery status.

Sum = 55.

Result:

The above program was executed and output was verified.